

PHÂN TÍCH THỐNG KÊ NHIỀU CHIỀU (MAT3452)

Bài tập tuần 11

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```
library(rattle)

## Warning: package 'rattle' was built under R version 4.5.3
## Loading required package: tibble
## Loading required package: bitops
## Rattle: A free graphical interface for data science with R.
## Version 5.6.2 Copyright (c) 2006-2023 Togaware Pty Ltd.
## Type 'rattle()' to shake, rattle, and roll your data.

data(wine)
head(wine)

##   Type Alcohol Malic  Ash Alkalinity Magnesium Phenols Flavanoids Nonflavanoids
## 1      1   14.23  1.71 2.43      15.6      127    2.80      3.06      0.28
## 2      1   13.20  1.78 2.14      11.2      100    2.65      2.76      0.26
## 3      1   13.16  2.36 2.67      18.6      101    2.80      3.24      0.30
## 4      1   14.37  1.95 2.50      16.8      113    3.85      3.49      0.24
## 5      1   13.24  2.59 2.87      21.0      118    2.80      2.69      0.39
## 6      1   14.20  1.76 2.45      15.2      112    3.27      3.39      0.34
##   Proanthocyanins Color  Hue Dilution Proline
## 1              2.29  5.64 1.04      3.92   1065
## 2              1.28  4.38 1.05      3.40   1050
## 3              2.81  5.68 1.03      3.17   1185
## 4              2.18  7.80 0.86      3.45   1480
## 5              1.82  4.32 1.04      2.93    735
## 6              1.97  6.75 1.05      2.85   1450

# Chuyển biến Type sang dạng số
wine$Type <- as.numeric(wine$Type)
table(wine$Type)

##
##  1  2  3
## 59 71 48

# Chia dữ liệu thành 3 nhóm theo từng loại rượu (Type = 1, 2, 3)
groups <- split(wine[, -1], wine$Type)
groups

## $`1`
##   Alcohol Malic  Ash Alkalinity Magnesium Phenols Flavanoids Nonflavanoids
## 1   14.23  1.71 2.43      15.6      127    2.80      3.06      0.28
## 2   13.20  1.78 2.14      11.2      100    2.65      2.76      0.26
## 3   13.16  2.36 2.67      18.6      101    2.80      3.24      0.30
## 4   14.37  1.95 2.50      16.8      113    3.85      3.49      0.24
## 5   13.24  2.59 2.87      21.0      118    2.80      2.69      0.39
## 6   14.20  1.76 2.45      15.2      112    3.27      3.39      0.34
## 7   14.39  1.87 2.45      14.6       96    2.50      2.52      0.30
## 8   14.06  2.15 2.61      17.6      121    2.60      2.51      0.31
## 9   14.83  1.64 2.17      14.0       97    2.80      2.98      0.29
## 10  13.86  1.35 2.27      16.0       98    2.98      3.15      0.22
```

## 11	14.10	2.16	2.30	18.0	105	2.95	3.32	0.22
## 12	14.12	1.48	2.32	16.8	95	2.20	2.43	0.26
## 13	13.75	1.73	2.41	16.0	89	2.60	2.76	0.29
## 14	14.75	1.73	2.39	11.4	91	3.10	3.69	0.43
## 15	14.38	1.87	2.38	12.0	102	3.30	3.64	0.29
## 16	13.63	1.81	2.70	17.2	112	2.85	2.91	0.30
## 17	14.30	1.92	2.72	20.0	120	2.80	3.14	0.33
## 18	13.83	1.57	2.62	20.0	115	2.95	3.40	0.40
## 19	14.19	1.59	2.48	16.5	108	3.30	3.93	0.32
## 20	13.64	3.10	2.56	15.2	116	2.70	3.03	0.17
## 21	14.06	1.63	2.28	16.0	126	3.00	3.17	0.24
## 22	12.93	3.80	2.65	18.6	102	2.41	2.41	0.25
## 23	13.71	1.86	2.36	16.6	101	2.61	2.88	0.27
## 24	12.85	1.60	2.52	17.8	95	2.48	2.37	0.26
## 25	13.50	1.81	2.61	20.0	96	2.53	2.61	0.28
## 26	13.05	2.05	3.22	25.0	124	2.63	2.68	0.47
## 27	13.39	1.77	2.62	16.1	93	2.85	2.94	0.34
## 28	13.30	1.72	2.14	17.0	94	2.40	2.19	0.27
## 29	13.87	1.90	2.80	19.4	107	2.95	2.97	0.37
## 30	14.02	1.68	2.21	16.0	96	2.65	2.33	0.26
## 31	13.73	1.50	2.70	22.5	101	3.00	3.25	0.29
## 32	13.58	1.66	2.36	19.1	106	2.86	3.19	0.22
## 33	13.68	1.83	2.36	17.2	104	2.42	2.69	0.42
## 34	13.76	1.53	2.70	19.5	132	2.95	2.74	0.50
## 35	13.51	1.80	2.65	19.0	110	2.35	2.53	0.29
## 36	13.48	1.81	2.41	20.5	100	2.70	2.98	0.26
## 37	13.28	1.64	2.84	15.5	110	2.60	2.68	0.34
## 38	13.05	1.65	2.55	18.0	98	2.45	2.43	0.29
## 39	13.07	1.50	2.10	15.5	98	2.40	2.64	0.28
## 40	14.22	3.99	2.51	13.2	128	3.00	3.04	0.20
## 41	13.56	1.71	2.31	16.2	117	3.15	3.29	0.34
## 42	13.41	3.84	2.12	18.8	90	2.45	2.68	0.27
## 43	13.88	1.89	2.59	15.0	101	3.25	3.56	0.17
## 44	13.24	3.98	2.29	17.5	103	2.64	2.63	0.32
## 45	13.05	1.77	2.10	17.0	107	3.00	3.00	0.28
## 46	14.21	4.04	2.44	18.9	111	2.85	2.65	0.30
## 47	14.38	3.59	2.28	16.0	102	3.25	3.17	0.27
## 48	13.90	1.68	2.12	16.0	101	3.10	3.39	0.21
## 49	14.10	2.02	2.40	18.8	103	2.75	2.92	0.32
## 50	13.94	1.73	2.27	17.4	108	2.88	3.54	0.32
## 51	13.05	1.73	2.04	12.4	92	2.72	3.27	0.17
## 52	13.83	1.65	2.60	17.2	94	2.45	2.99	0.22
## 53	13.82	1.75	2.42	14.0	111	3.88	3.74	0.32
## 54	13.77	1.90	2.68	17.1	115	3.00	2.79	0.39
## 55	13.74	1.67	2.25	16.4	118	2.60	2.90	0.21
## 56	13.56	1.73	2.46	20.5	116	2.96	2.78	0.20
## 57	14.22	1.70	2.30	16.3	118	3.20	3.00	0.26
## 58	13.29	1.97	2.68	16.8	102	3.00	3.23	0.31
## 59	13.72	1.43	2.50	16.7	108	3.40	3.67	0.19
##	Proanthocyanins Color			Hue	Dilution	Proline		
## 1		2.29	5.64	1.04	3.92	1065		
## 2		1.28	4.38	1.05	3.40	1050		
## 3		2.81	5.68	1.03	3.17	1185		
## 4		2.18	7.80	0.86	3.45	1480		
## 5		1.82	4.32	1.04	2.93	735		
## 6		1.97	6.75	1.05	2.85	1450		

## 7	1.98	5.25	1.02	3.58	1290
## 8	1.25	5.05	1.06	3.58	1295
## 9	1.98	5.20	1.08	2.85	1045
## 10	1.85	7.22	1.01	3.55	1045
## 11	2.38	5.75	1.25	3.17	1510
## 12	1.57	5.00	1.17	2.82	1280
## 13	1.81	5.60	1.15	2.90	1320
## 14	2.81	5.40	1.25	2.73	1150
## 15	2.96	7.50	1.20	3.00	1547
## 16	1.46	7.30	1.28	2.88	1310
## 17	1.97	6.20	1.07	2.65	1280
## 18	1.72	6.60	1.13	2.57	1130
## 19	1.86	8.70	1.23	2.82	1680
## 20	1.66	5.10	0.96	3.36	845
## 21	2.10	5.65	1.09	3.71	780
## 22	1.98	4.50	1.03	3.52	770
## 23	1.69	3.80	1.11	4.00	1035
## 24	1.46	3.93	1.09	3.63	1015
## 25	1.66	3.52	1.12	3.82	845
## 26	1.92	3.58	1.13	3.20	830
## 27	1.45	4.80	0.92	3.22	1195
## 28	1.35	3.95	1.02	2.77	1285
## 29	1.76	4.50	1.25	3.40	915
## 30	1.98	4.70	1.04	3.59	1035
## 31	2.38	5.70	1.19	2.71	1285
## 32	1.95	6.90	1.09	2.88	1515
## 33	1.97	3.84	1.23	2.87	990
## 34	1.35	5.40	1.25	3.00	1235
## 35	1.54	4.20	1.10	2.87	1095
## 36	1.86	5.10	1.04	3.47	920
## 37	1.36	4.60	1.09	2.78	880
## 38	1.44	4.25	1.12	2.51	1105
## 39	1.37	3.70	1.18	2.69	1020
## 40	2.08	5.10	0.89	3.53	760
## 41	2.34	6.13	0.95	3.38	795
## 42	1.48	4.28	0.91	3.00	1035
## 43	1.70	5.43	0.88	3.56	1095
## 44	1.66	4.36	0.82	3.00	680
## 45	2.03	5.04	0.88	3.35	885
## 46	1.25	5.24	0.87	3.33	1080
## 47	2.19	4.90	1.04	3.44	1065
## 48	2.14	6.10	0.91	3.33	985
## 49	2.38	6.20	1.07	2.75	1060
## 50	2.08	8.90	1.12	3.10	1260
## 51	2.91	7.20	1.12	2.91	1150
## 52	2.29	5.60	1.24	3.37	1265
## 53	1.87	7.05	1.01	3.26	1190
## 54	1.68	6.30	1.13	2.93	1375
## 55	1.62	5.85	0.92	3.20	1060
## 56	2.45	6.25	0.98	3.03	1120
## 57	2.03	6.38	0.94	3.31	970
## 58	1.66	6.00	1.07	2.84	1270
## 59	2.04	6.80	0.89	2.87	1285
##					
## \$`2`					
##	Alcohol	Malic	Ash	Alcalinity	Magnesium Phenols Flavanoids Nonflavanoids

## 60	12.37	0.94	1.36	10.6	88	1.98	0.57	0.28
## 61	12.33	1.10	2.28	16.0	101	2.05	1.09	0.63
## 62	12.64	1.36	2.02	16.8	100	2.02	1.41	0.53
## 63	13.67	1.25	1.92	18.0	94	2.10	1.79	0.32
## 64	12.37	1.13	2.16	19.0	87	3.50	3.10	0.19
## 65	12.17	1.45	2.53	19.0	104	1.89	1.75	0.45
## 66	12.37	1.21	2.56	18.1	98	2.42	2.65	0.37
## 67	13.11	1.01	1.70	15.0	78	2.98	3.18	0.26
## 68	12.37	1.17	1.92	19.6	78	2.11	2.00	0.27
## 69	13.34	0.94	2.36	17.0	110	2.53	1.30	0.55
## 70	12.21	1.19	1.75	16.8	151	1.85	1.28	0.14
## 71	12.29	1.61	2.21	20.4	103	1.10	1.02	0.37
## 72	13.86	1.51	2.67	25.0	86	2.95	2.86	0.21
## 73	13.49	1.66	2.24	24.0	87	1.88	1.84	0.27
## 74	12.99	1.67	2.60	30.0	139	3.30	2.89	0.21
## 75	11.96	1.09	2.30	21.0	101	3.38	2.14	0.13
## 76	11.66	1.88	1.92	16.0	97	1.61	1.57	0.34
## 77	13.03	0.90	1.71	16.0	86	1.95	2.03	0.24
## 78	11.84	2.89	2.23	18.0	112	1.72	1.32	0.43
## 79	12.33	0.99	1.95	14.8	136	1.90	1.85	0.35
## 80	12.70	3.87	2.40	23.0	101	2.83	2.55	0.43
## 81	12.00	0.92	2.00	19.0	86	2.42	2.26	0.30
## 82	12.72	1.81	2.20	18.8	86	2.20	2.53	0.26
## 83	12.08	1.13	2.51	24.0	78	2.00	1.58	0.40
## 84	13.05	3.86	2.32	22.5	85	1.65	1.59	0.61
## 85	11.84	0.89	2.58	18.0	94	2.20	2.21	0.22
## 86	12.67	0.98	2.24	18.0	99	2.20	1.94	0.30
## 87	12.16	1.61	2.31	22.8	90	1.78	1.69	0.43
## 88	11.65	1.67	2.62	26.0	88	1.92	1.61	0.40
## 89	11.64	2.06	2.46	21.6	84	1.95	1.69	0.48
## 90	12.08	1.33	2.30	23.6	70	2.20	1.59	0.42
## 91	12.08	1.83	2.32	18.5	81	1.60	1.50	0.52
## 92	12.00	1.51	2.42	22.0	86	1.45	1.25	0.50
## 93	12.69	1.53	2.26	20.7	80	1.38	1.46	0.58
## 94	12.29	2.83	2.22	18.0	88	2.45	2.25	0.25
## 95	11.62	1.99	2.28	18.0	98	3.02	2.26	0.17
## 96	12.47	1.52	2.20	19.0	162	2.50	2.27	0.32
## 97	11.81	2.12	2.74	21.5	134	1.60	0.99	0.14
## 98	12.29	1.41	1.98	16.0	85	2.55	2.50	0.29
## 99	12.37	1.07	2.10	18.5	88	3.52	3.75	0.24
## 100	12.29	3.17	2.21	18.0	88	2.85	2.99	0.45
## 101	12.08	2.08	1.70	17.5	97	2.23	2.17	0.26
## 102	12.60	1.34	1.90	18.5	88	1.45	1.36	0.29
## 103	12.34	2.45	2.46	21.0	98	2.56	2.11	0.34
## 104	11.82	1.72	1.88	19.5	86	2.50	1.64	0.37
## 105	12.51	1.73	1.98	20.5	85	2.20	1.92	0.32
## 106	12.42	2.55	2.27	22.0	90	1.68	1.84	0.66
## 107	12.25	1.73	2.12	19.0	80	1.65	2.03	0.37
## 108	12.72	1.75	2.28	22.5	84	1.38	1.76	0.48
## 109	12.22	1.29	1.94	19.0	92	2.36	2.04	0.39
## 110	11.61	1.35	2.70	20.0	94	2.74	2.92	0.29
## 111	11.46	3.74	1.82	19.5	107	3.18	2.58	0.24
## 112	12.52	2.43	2.17	21.0	88	2.55	2.27	0.26
## 113	11.76	2.68	2.92	20.0	103	1.75	2.03	0.60
## 114	11.41	0.74	2.50	21.0	88	2.48	2.01	0.42
## 115	12.08	1.39	2.50	22.5	84	2.56	2.29	0.43

## 116	11.03	1.51	2.20	21.5	85	2.46	2.17	0.52
## 117	11.82	1.47	1.99	20.8	86	1.98	1.60	0.30
## 118	12.42	1.61	2.19	22.5	108	2.00	2.09	0.34
## 119	12.77	3.43	1.98	16.0	80	1.63	1.25	0.43
## 120	12.00	3.43	2.00	19.0	87	2.00	1.64	0.37
## 121	11.45	2.40	2.42	20.0	96	2.90	2.79	0.32
## 122	11.56	2.05	3.23	28.5	119	3.18	5.08	0.47
## 123	12.42	4.43	2.73	26.5	102	2.20	2.13	0.43
## 124	13.05	5.80	2.13	21.5	86	2.62	2.65	0.30
## 125	11.87	4.31	2.39	21.0	82	2.86	3.03	0.21
## 126	12.07	2.16	2.17	21.0	85	2.60	2.65	0.37
## 127	12.43	1.53	2.29	21.5	86	2.74	3.15	0.39
## 128	11.79	2.13	2.78	28.5	92	2.13	2.24	0.58
## 129	12.37	1.63	2.30	24.5	88	2.22	2.45	0.40
## 130	12.04	4.30	2.38	22.0	80	2.10	1.75	0.42
##	Proanthocyanins Color			Hue	Dilution	Proline		
## 60		0.42	1.95	1.050	1.82	520		
## 61		0.41	3.27	1.250	1.67	680		
## 62		0.62	5.75	0.980	1.59	450		
## 63		0.73	3.80	1.230	2.46	630		
## 64		1.87	4.45	1.220	2.87	420		
## 65		1.03	2.95	1.450	2.23	355		
## 66		2.08	4.60	1.190	2.30	678		
## 67		2.28	5.30	1.120	3.18	502		
## 68		1.04	4.68	1.120	3.48	510		
## 69		0.42	3.17	1.020	1.93	750		
## 70		2.50	2.85	1.280	3.07	718		
## 71		1.46	3.05	0.906	1.82	870		
## 72		1.87	3.38	1.360	3.16	410		
## 73		1.03	3.74	0.980	2.78	472		
## 74		1.96	3.35	1.310	3.50	985		
## 75		1.65	3.21	0.990	3.13	886		
## 76		1.15	3.80	1.230	2.14	428		
## 77		1.46	4.60	1.190	2.48	392		
## 78		0.95	2.65	0.960	2.52	500		
## 79		2.76	3.40	1.060	2.31	750		
## 80		1.95	2.57	1.190	3.13	463		
## 81		1.43	2.50	1.380	3.12	278		
## 82		1.77	3.90	1.160	3.14	714		
## 83		1.40	2.20	1.310	2.72	630		
## 84		1.62	4.80	0.840	2.01	515		
## 85		2.35	3.05	0.790	3.08	520		
## 86		1.46	2.62	1.230	3.16	450		
## 87		1.56	2.45	1.330	2.26	495		
## 88		1.34	2.60	1.360	3.21	562		
## 89		1.35	2.80	1.000	2.75	680		
## 90		1.38	1.74	1.070	3.21	625		
## 91		1.64	2.40	1.080	2.27	480		
## 92		1.63	3.60	1.050	2.65	450		
## 93		1.62	3.05	0.960	2.06	495		
## 94		1.99	2.15	1.150	3.30	290		
## 95		1.35	3.25	1.160	2.96	345		
## 96		3.28	2.60	1.160	2.63	937		
## 97		1.56	2.50	0.950	2.26	625		
## 98		1.77	2.90	1.230	2.74	428		
## 99		1.95	4.50	1.040	2.77	660		

## 100	2.81	2.30	1.420	2.83	406
## 101	1.40	3.30	1.270	2.96	710
## 102	1.35	2.45	1.040	2.77	562
## 103	1.31	2.80	0.800	3.38	438
## 104	1.42	2.06	0.940	2.44	415
## 105	1.48	2.94	1.040	3.57	672
## 106	1.42	2.70	0.860	3.30	315
## 107	1.63	3.40	1.000	3.17	510
## 108	1.63	3.30	0.880	2.42	488
## 109	2.08	2.70	0.860	3.02	312
## 110	2.49	2.65	0.960	3.26	680
## 111	3.58	2.90	0.750	2.81	562
## 112	1.22	2.00	0.900	2.78	325
## 113	1.05	3.80	1.230	2.50	607
## 114	1.44	3.08	1.100	2.31	434
## 115	1.04	2.90	0.930	3.19	385
## 116	2.01	1.90	1.710	2.87	407
## 117	1.53	1.95	0.950	3.33	495
## 118	1.61	2.06	1.060	2.96	345
## 119	0.83	3.40	0.700	2.12	372
## 120	1.87	1.28	0.930	3.05	564
## 121	1.83	3.25	0.800	3.39	625
## 122	1.87	6.00	0.930	3.69	465
## 123	1.71	2.08	0.920	3.12	365
## 124	2.01	2.60	0.730	3.10	380
## 125	2.91	2.80	0.750	3.64	380
## 126	1.35	2.76	0.860	3.28	378
## 127	1.77	3.94	0.690	2.84	352
## 128	1.76	3.00	0.970	2.44	466
## 129	1.90	2.12	0.890	2.78	342
## 130	1.35	2.60	0.790	2.57	580

##

\$`3`

##	Alcohol	Malic	Ash	Alcalinity	Magnesium	Phenols	Flavanoids	Nonflavanoids
## 131	12.86	1.35	2.32	18.0	122	1.51	1.25	0.21
## 132	12.88	2.99	2.40	20.0	104	1.30	1.22	0.24
## 133	12.81	2.31	2.40	24.0	98	1.15	1.09	0.27
## 134	12.70	3.55	2.36	21.5	106	1.70	1.20	0.17
## 135	12.51	1.24	2.25	17.5	85	2.00	0.58	0.60
## 136	12.60	2.46	2.20	18.5	94	1.62	0.66	0.63
## 137	12.25	4.72	2.54	21.0	89	1.38	0.47	0.53
## 138	12.53	5.51	2.64	25.0	96	1.79	0.60	0.63
## 139	13.49	3.59	2.19	19.5	88	1.62	0.48	0.58
## 140	12.84	2.96	2.61	24.0	101	2.32	0.60	0.53
## 141	12.93	2.81	2.70	21.0	96	1.54	0.50	0.53
## 142	13.36	2.56	2.35	20.0	89	1.40	0.50	0.37
## 143	13.52	3.17	2.72	23.5	97	1.55	0.52	0.50
## 144	13.62	4.95	2.35	20.0	92	2.00	0.80	0.47
## 145	12.25	3.88	2.20	18.5	112	1.38	0.78	0.29
## 146	13.16	3.57	2.15	21.0	102	1.50	0.55	0.43
## 147	13.88	5.04	2.23	20.0	80	0.98	0.34	0.40
## 148	12.87	4.61	2.48	21.5	86	1.70	0.65	0.47
## 149	13.32	3.24	2.38	21.5	92	1.93	0.76	0.45
## 150	13.08	3.90	2.36	21.5	113	1.41	1.39	0.34
## 151	13.50	3.12	2.62	24.0	123	1.40	1.57	0.22
## 152	12.79	2.67	2.48	22.0	112	1.48	1.36	0.24

## 153	13.11	1.90	2.75	25.5	116	2.20	1.28	0.26
## 154	13.23	3.30	2.28	18.5	98	1.80	0.83	0.61
## 155	12.58	1.29	2.10	20.0	103	1.48	0.58	0.53
## 156	13.17	5.19	2.32	22.0	93	1.74	0.63	0.61
## 157	13.84	4.12	2.38	19.5	89	1.80	0.83	0.48
## 158	12.45	3.03	2.64	27.0	97	1.90	0.58	0.63
## 159	14.34	1.68	2.70	25.0	98	2.80	1.31	0.53
## 160	13.48	1.67	2.64	22.5	89	2.60	1.10	0.52
## 161	12.36	3.83	2.38	21.0	88	2.30	0.92	0.50
## 162	13.69	3.26	2.54	20.0	107	1.83	0.56	0.50
## 163	12.85	3.27	2.58	22.0	106	1.65	0.60	0.60
## 164	12.96	3.45	2.35	18.5	106	1.39	0.70	0.40
## 165	13.78	2.76	2.30	22.0	90	1.35	0.68	0.41
## 166	13.73	4.36	2.26	22.5	88	1.28	0.47	0.52
## 167	13.45	3.70	2.60	23.0	111	1.70	0.92	0.43
## 168	12.82	3.37	2.30	19.5	88	1.48	0.66	0.40
## 169	13.58	2.58	2.69	24.5	105	1.55	0.84	0.39
## 170	13.40	4.60	2.86	25.0	112	1.98	0.96	0.27
## 171	12.20	3.03	2.32	19.0	96	1.25	0.49	0.40
## 172	12.77	2.39	2.28	19.5	86	1.39	0.51	0.48
## 173	14.16	2.51	2.48	20.0	91	1.68	0.70	0.44
## 174	13.71	5.65	2.45	20.5	95	1.68	0.61	0.52
## 175	13.40	3.91	2.48	23.0	102	1.80	0.75	0.43
## 176	13.27	4.28	2.26	20.0	120	1.59	0.69	0.43
## 177	13.17	2.59	2.37	20.0	120	1.65	0.68	0.53
## 178	14.13	4.10	2.74	24.5	96	2.05	0.76	0.56
##	Proanthocyanins			Color	Hue	Dilution	Proline	
## 131		0.94	4.100000	0.76		1.29	630	
## 132		0.83	5.400000	0.74		1.42	530	
## 133		0.83	5.700000	0.66		1.36	560	
## 134		0.84	5.000000	0.78		1.29	600	
## 135		1.25	5.450000	0.75		1.51	650	
## 136		0.94	7.100000	0.73		1.58	695	
## 137		0.80	3.850000	0.75		1.27	720	
## 138		1.10	5.000000	0.82		1.69	515	
## 139		0.88	5.700000	0.81		1.82	580	
## 140		0.81	4.920000	0.89		2.15	590	
## 141		0.75	4.600000	0.77		2.31	600	
## 142		0.64	5.600000	0.70		2.47	780	
## 143		0.55	4.350000	0.89		2.06	520	
## 144		1.02	4.400000	0.91		2.05	550	
## 145		1.14	8.210000	0.65		2.00	855	
## 146		1.30	4.000000	0.60		1.68	830	
## 147		0.68	4.900000	0.58		1.33	415	
## 148		0.86	7.650000	0.54		1.86	625	
## 149		1.25	8.420000	0.55		1.62	650	
## 150		1.14	9.400000	0.57		1.33	550	
## 151		1.25	8.600000	0.59		1.30	500	
## 152		1.26	10.800000	0.48		1.47	480	
## 153		1.56	7.100000	0.61		1.33	425	
## 154		1.87	10.520000	0.56		1.51	675	
## 155		1.40	7.600000	0.58		1.55	640	
## 156		1.55	7.900000	0.60		1.48	725	
## 157		1.56	9.010000	0.57		1.64	480	
## 158		1.14	7.500000	0.67		1.73	880	
## 159		2.70	13.000000	0.57		1.96	660	

```
## 160      2.29 11.750000 0.57      1.78      620
## 161      1.04  7.650000 0.56      1.58      520
## 162      0.80  5.880000 0.96      1.82      680
## 163      0.96  5.580000 0.87      2.11      570
## 164      0.94  5.280000 0.68      1.75      675
## 165      1.03  9.580000 0.70      1.68      615
## 166      1.15  6.620000 0.78      1.75      520
## 167      1.46 10.680000 0.85      1.56      695
## 168      0.97 10.260000 0.72      1.75      685
## 169      1.54  8.660000 0.74      1.80      750
## 170      1.11  8.500000 0.67      1.92      630
## 171      0.73  5.500000 0.66      1.83      510
## 172      0.64  9.899999 0.57      1.63      470
## 173      1.24  9.700000 0.62      1.71      660
## 174      1.06  7.700000 0.64      1.74      740
## 175      1.41  7.300000 0.70      1.56      750
## 176      1.35 10.200000 0.59      1.56      835
## 177      1.46  9.300000 0.60      1.62      840
## 178      1.35  9.200000 0.61      1.60      560
```

```
# Để tính Si(U), cần tìm GTTB của từng nhóm, ma trận hiệp phương sai chung A,
# Tỷ lệ i của các nhóm và áp dụng trên chai rượu có giá trị U để dự đoán.
# Dự đoán trên tất cả các chai rượu của bộ dữ liệu wine ban đầu.
# A = (i=1_k) (ni - 1)Ai / (i=1_k) (ni - 1), Ai là ma trận cov của nhóm i.
# Tính số quan sát trong mỗi nhóm dữ liệu
n <- sapply(groups, nrow)
# Tính tỷ lệ i của từng nhóm
pi <- n / length(wine$Type)
# Tính GTTB μi của từng nhóm
mu <- sapply(groups, colMeans)
# Tính ma trận hiệp phương sai của từng nhóm
Ai <- lapply(groups, cov) # nolint
# Tính ma trận hiệp phương sai gộp
A <- 0 # nolint
for (i in 1:3) {
  A <- A + (n[i] - 1) * Ai[[as.character(i)]] # nolint
}
A <- A / sum(n - 1) # nolint
round(A, 3)
```

```
##      Alcohol      Malic      Ash Alcalinity Magnesium Phenols Flavanoids
## Alcohol      0.262      0.008 -0.013      -0.097      0.018      0.027      0.023
## Malic         0.008      0.888      0.021      0.415     -0.889     -0.014     -0.009
## Ash          -0.013      0.021      0.066      0.484      0.690      0.016      0.030
## Alcalinity   -0.097      0.415      0.484      8.007      3.238      0.109      0.246
## Magnesium     0.018     -0.889      0.690      3.238     180.658      0.577      0.667
## Phenols       0.027     -0.014      0.016      0.109      0.577      0.191      0.161
## Flavanoids    0.023     -0.009      0.030      0.246      0.667      0.161      0.275
## Nonflavanoids -0.001      0.010      0.007      0.047     -0.287     -0.008     -0.015
## Proanthocyanins 0.017      0.017      0.003      0.093      1.300      0.092      0.128
## Color         0.246     -0.259      0.010     -0.102      1.772      0.199      0.287
## Hue           0.001     -0.042      0.002     -0.010      0.126     -0.001     -0.004
## Dilution     -0.005      0.047      0.011      0.226     -0.281      0.060      0.068
## Proline       12.237    -33.058     -0.501     -32.833     476.249      8.395      3.426
##
## Nonflavanoids Proanthocyanins      Color      Hue Dilution      Proline
## Alcohol         -0.001              0.017      0.246      0.001     -0.005      12.237
## Malic            0.010              0.017     -0.259     -0.042      0.047     -33.058
```



```
## Ash                0.007                0.003  0.010  0.002    0.011    -0.501
## Alcalinity         0.047                0.093 -0.102 -0.010    0.226   -32.833
## Magnesium         -0.287                1.300  1.772  0.126   -0.281   476.249
## Phenols           -0.008                0.092  0.199 -0.001    0.060    8.395
## Flavanoids        -0.015                0.128  0.287 -0.004    0.068    3.426
## Nonflavanoids     0.012               -0.009 -0.002  0.001   -0.010   -0.493
## Proanthocyanins   -0.009                0.246  0.229 -0.006    0.042   11.482
## Color             -0.002                0.229  2.285 -0.041   -0.066   68.094
## Hue                0.001               -0.006 -0.041  0.024   -0.003    4.491
## Dilution         -0.010                0.042 -0.066 -0.003    0.161  -10.943
## Proline           -0.493               11.482 68.094  4.491  -10.943 29707.682
```

```
# Tính hàm thông tin phân biệt cho từng nhóm
S <- list() # nolint
for (i in 1:3) {
  term1 <- t(mu[, i]) %*% solve(A) %*% t(wine[, -1])
  term2 <- 0.5 * t(mu[, i]) %*% solve(A) %*% mu[, i]
  Si <- term1 - as.numeric(term2) + log(pi[i]) # nolint
  S[[i]] <- t(Si) # nolint
}
S <- as.data.frame(S) # nolint
S
```

```
##      structure.c.574.687156773102..484.371231692448..495.455706392936..
## 1                                     574.6872
## 2                                     484.3712
## 3                                     495.4557
## 4                                     558.6600
## 5                                     506.3195
## 6                                     547.8827
## 7                                     567.3543
## 8                                     566.5965
## 9                                     562.9682
## 10                                    513.0377
## 11                                    545.7767
## 12                                    530.8747
## 13                                    511.9514
## 14                                    577.2650
## 15                                    559.2552
## 16                                    523.0308
## 17                                    560.0858
## 18                                    524.4761
## 19                                    547.6549
## 20                                    519.7067
## 21                                    546.8722
## 22                                    482.4944
## 23                                    537.8419
## 24                                    478.8357
## 25                                    523.8086
## 26                                    526.2823
## 27                                    504.2994
## 28                                    474.0282
## 29                                    553.7138
## 30                                    530.6103
## 31                                    520.4992
## 32                                    501.5047
## 33                                    520.1988
```

## 34	553.5076
## 35	514.6818
## 36	501.9168
## 37	504.2134
## 38	470.5544
## 39	462.2389
## 40	560.0260
## 41	506.0894
## 42	475.5688
## 43	532.5333
## 44	471.2023
## 45	463.4421
## 46	547.5019
## 47	556.7803
## 48	507.2024
## 49	525.6626
## 50	518.6626
## 51	448.7921
## 52	536.6674
## 53	525.0977
## 54	537.1695
## 55	510.4819
## 56	501.1951
## 57	539.9978
## 58	497.0962
## 59	506.1621
## 60	357.1735
## 61	409.5507
## 62	390.0036
## 63	472.1939
## 64	401.1345
## 65	419.1479
## 66	423.3124
## 67	431.4507
## 68	401.5537
## 69	471.7004
## 70	425.9473
## 71	395.5981
## 72	518.5279
## 73	463.7891
## 74	506.6984
## 75	397.1627
## 76	351.0149
## 77	414.6156
## 78	389.7495
## 79	420.0650
## 80	459.3831
## 81	392.1981
## 82	440.5404
## 83	409.9570
## 84	433.0603
## 85	394.4571
## 86	446.6804
## 87	402.8554
## 88	403.4491
## 89	382.5908

## 90	404.4803
## 91	393.9655
## 92	394.3123
## 93	416.9966
## 94	418.6449
## 95	370.4409
## 96	464.6012
## 97	401.5262
## 98	399.2265
## 99	401.4687
## 100	427.5027
## 101	390.3728
## 102	412.6606
## 103	428.0629
## 104	354.9557
## 105	427.3813
## 106	433.0702
## 107	405.4686
## 108	422.3831
## 109	392.8836
## 110	402.4036
## 111	345.6909
## 112	409.4503
## 113	420.0119
## 114	356.0233
## 115	407.4790
## 116	360.2481
## 117	379.5119
## 118	426.2835
## 119	397.1284
## 120	394.1082
## 121	378.5594
## 122	431.4316
## 123	447.3663
## 124	444.7898
## 125	394.3653
## 126	395.8515
## 127	403.6135
## 128	395.9971
## 129	410.6322
## 130	388.2725
## 131	410.8486
## 132	402.9155
## 133	389.7411
## 134	388.6026
## 135	381.2785
## 136	387.0876
## 137	381.0645
## 138	408.5858
## 139	443.1478
## 140	434.3035
## 141	446.2154
## 142	449.5474
## 143	475.6079
## 144	467.9095
## 145	371.3841

## 146	426.9017
## 147	440.5601
## 148	400.0212
## 149	415.4188
## 150	399.4627
## 151	434.4816
## 152	375.1193
## 153	415.6230
## 154	409.1274
## 155	373.3894
## 156	415.0303
## 157	443.1879
## 158	395.0321
## 159	480.0250
## 160	424.5898
## 161	360.0282
## 162	479.0217
## 163	438.4481
## 164	421.5246
## 165	436.8208
## 166	450.0290
## 167	440.0289
## 168	382.8376
## 169	456.3221
## 170	450.1985
## 171	367.4596
## 172	370.0040
## 173	466.6196
## 174	457.4812
## 175	437.6253
## 176	420.4683
## 177	426.7725
## 178	477.0003
## structure.c.555.146118953013..469.529368913997..482.482350282507..	
## 1	555.1461
## 2	469.5294
## 3	482.4824
## 4	532.3444
## 5	503.8055
## 6	523.8098
## 7	543.0192
## 8	544.2155
## 9	546.3404
## 10	499.2060
## 11	524.8156
## 12	514.7701
## 13	494.6873
## 14	554.4405
## 15	530.2323
## 16	504.7645
## 17	541.9849
## 18	512.9321
## 19	519.0810
## 20	505.4021
## 21	536.9839
## 22	477.4513

## 23	522.2823
## 24	470.3285
## 25	515.8747
## 26	522.8436
## 27	486.3416
## 28	464.3881
## 29	541.8925
## 30	518.6782
## 31	509.3815
## 32	484.2408
## 33	512.4579
## 34	538.9828
## 35	503.4365
## 36	495.9985
## 37	492.8696
## 38	463.4811
## 39	457.5044
## 40	542.6601
## 41	499.6761
## 42	468.6740
## 43	510.5720
## 44	469.7422
## 45	460.6823
## 46	531.2995
## 47	539.6496
## 48	496.1195
## 49	516.2798
## 50	502.6383
## 51	438.7853
## 52	517.4169
## 53	505.4648
## 54	517.2946
## 55	497.8196
## 56	494.0438
## 57	527.6001
## 58	479.8772
## 59	487.8323
## 60	376.9820
## 61	422.4417
## 62	408.4186
## 63	480.0289
## 64	417.0382
## 65	436.4536
## 66	430.4241
## 67	440.5435
## 68	415.6029
## 69	476.7122
## 70	441.0074
## 71	407.7041
## 72	526.9466
## 73	476.3631
## 74	512.7032
## 75	406.6116
## 76	372.1120
## 77	430.7841
## 78	404.7490

## 79	430.9549
## 80	472.1722
## 81	414.1234
## 82	445.1146
## 83	426.9797
## 84	447.5659
## 85	403.8597
## 86	457.8044
## 87	424.1804
## 88	422.4811
## 89	396.3144
## 90	420.8521
## 91	410.5628
## 92	413.9751
## 93	433.8475
## 94	433.3185
## 95	388.7329
## 96	471.9920
## 97	415.2680
## 98	414.1503
## 99	410.2615
## 100	442.5860
## 101	403.7380
## 102	426.9632
## 103	438.5853
## 104	379.8262
## 105	437.7096
## 106	449.2516
## 107	417.5730
## 108	437.9239
## 109	413.8633
## 110	409.1016
## 111	368.0014
## 112	426.5139
## 113	427.2919
## 114	378.1234
## 115	422.8078
## 116	388.4288
## 117	398.5953
## 118	445.4646
## 119	411.0899
## 120	408.9005
## 121	388.2150
## 122	437.2108
## 123	462.1824
## 124	455.0799
## 125	406.6822
## 126	411.3754
## 127	418.2404
## 128	418.1451
## 129	431.2259
## 130	401.8026
## 131	421.9899
## 132	415.4112
## 133	407.1658
## 134	403.7133

## 135	398.1316
## 136	401.5680
## 137	394.8671
## 138	427.9083
## 139	455.6126
## 140	450.3753
## 141	454.7994
## 142	453.6174
## 143	486.2272
## 144	477.1631
## 145	380.0837
## 146	437.4038
## 147	453.1371
## 148	411.4512
## 149	428.0160
## 150	412.6230
## 151	446.4698
## 152	390.8971
## 153	435.4275
## 154	420.8840
## 155	392.9420
## 156	428.1483
## 157	453.9999
## 158	411.1776
## 159	490.7730
## 160	438.4916
## 161	379.2300
## 162	485.7127
## 163	451.8037
## 164	430.0320
## 165	448.6559
## 166	464.5545
## 167	450.7550
## 168	394.8556
## 169	465.1668
## 170	458.4540
## 171	384.1405
## 172	387.5017
## 173	472.2427
## 174	462.1208
## 175	448.4148
## 176	428.4432
## 177	435.3541
## 178	487.2572
## structure.c.534.532917172422..447.394437132448..465.276690858956..	
## 1	534.5329
## 2	447.3944
## 3	465.2767
## 4	522.1069
## 5	491.0545
## 6	508.9949
## 7	528.3928
## 8	530.5901
## 9	529.3201
## 10	483.2494
## 11	504.7750

## 12	500.5540
## 13	479.7767
## 14	530.2226
## 15	512.7094
## 16	494.5140
## 17	530.7726
## 18	497.5717
## 19	503.0729
## 20	491.6671
## 21	515.5772
## 22	465.7762
## 23	496.7634
## 24	451.2654
## 25	494.5255
## 26	506.1316
## 27	469.6840
## 28	449.5658
## 29	522.4762
## 30	502.3561
## 31	493.6204
## 32	470.0760
## 33	490.4354
## 34	522.4685
## 35	488.9047
## 36	476.7275
## 37	479.1520
## 38	450.0556
## 39	435.2816
## 40	530.8731
## 41	479.9178
## 42	454.6166
## 43	492.3170
## 44	458.1068
## 45	439.9626
## 46	523.6476
## 47	522.9036
## 48	476.8526
## 49	503.8384
## 50	487.6355
## 51	420.2429
## 52	498.9565
## 53	487.5221
## 54	506.0485
## 55	482.5925
## 56	483.5648
## 57	514.0218
## 58	465.2994
## 59	473.4026
## 60	366.2216
## 61	412.3886
## 62	405.0924
## 63	465.2561
## 64	394.0485
## 65	419.0002
## 66	412.8071
## 67	413.0913

## 68	395.6416
## 69	469.7607
## 70	420.8209
## 71	401.3255
## 72	506.8744
## 73	465.4036
## 74	489.8522
## 75	391.1261
## 76	356.7031
## 77	412.8496
## 78	393.4171
## 79	411.3124
## 80	451.5357
## 81	384.8908
## 82	423.2153
## 83	408.5520
## 84	445.3987
## 85	386.3221
## 86	435.4335
## 87	405.6654
## 88	403.0150
## 89	380.5023
## 90	400.1452
## 91	394.4632
## 92	401.9072
## 93	420.8616
## 94	410.8841
## 95	371.0867
## 96	448.7169
## 97	413.5805
## 98	388.3004
## 99	382.2151
## 100	411.7362
## 101	378.9939
## 102	409.6133
## 103	423.0381
## 104	362.4436
## 105	415.3832
## 106	428.2476
## 107	396.1950
## 108	423.8830
## 109	391.5462
## 110	383.3418
## 111	348.7327
## 112	408.5941
## 113	412.6955
## 114	360.6336
## 115	402.5802
## 116	354.5378
## 117	376.8209
## 118	422.4710
## 119	407.6353
## 120	389.5430
## 121	366.1625
## 122	403.3731
## 123	448.9274

## 124	441.6922
## 125	385.7113
## 126	387.1218
## 127	396.4003
## 128	401.9250
## 129	408.3866
## 130	393.6047
## 131	424.7144
## 132	424.7726
## 133	417.7586
## 134	414.9840
## 135	404.5223
## 136	412.2952
## 137	407.8336
## 138	442.6210
## 139	467.0303
## 140	459.9607
## 141	463.2508
## 142	463.8221
## 143	496.5858
## 144	486.6492
## 145	396.9502
## 146	446.8581
## 147	473.3040
## 148	433.8305
## 149	450.2810
## 150	433.4683
## 151	466.3804
## 152	415.3578
## 153	453.3328
## 154	444.3528
## 155	404.8982
## 156	450.7815
## 157	478.4616
## 158	429.1200
## 159	519.6569
## 160	464.7464
## 161	398.0606
## 162	499.3001
## 163	460.8922
## 164	441.3956
## 165	471.3945
## 166	481.5863
## 167	477.0525
## 168	418.8063
## 169	485.8843
## 170	485.7063
## 171	395.4190
## 172	410.8073
## 173	497.4518
## 174	486.8278
## 175	468.7611
## 176	455.6936
## 177	456.1091
## 178	516.0440

```
colnames(S) <- paste0("Type", 1:3) # nolint
S
```

##	Type1	Type2	Type3
## 1	574.6872	555.1461	534.5329
## 2	484.3712	469.5294	447.3944
## 3	495.4557	482.4824	465.2767
## 4	558.6600	532.3444	522.1069
## 5	506.3195	503.8055	491.0545
## 6	547.8827	523.8098	508.9949
## 7	567.3543	543.0192	528.3928
## 8	566.5965	544.2155	530.5901
## 9	562.9682	546.3404	529.3201
## 10	513.0377	499.2060	483.2494
## 11	545.7767	524.8156	504.7750
## 12	530.8747	514.7701	500.5540
## 13	511.9514	494.6873	479.7767
## 14	577.2650	554.4405	530.2226
## 15	559.2552	530.2323	512.7094
## 16	523.0308	504.7645	494.5140
## 17	560.0858	541.9849	530.7726
## 18	524.4761	512.9321	497.5717
## 19	547.6549	519.0810	503.0729
## 20	519.7067	505.4021	491.6671
## 21	546.8722	536.9839	515.5772
## 22	482.4944	477.4513	465.7762
## 23	537.8419	522.2823	496.7634
## 24	478.8357	470.3285	451.2654
## 25	523.8086	515.8747	494.5255
## 26	526.2823	522.8436	506.1316
## 27	504.2994	486.3416	469.6840
## 28	474.0282	464.3881	449.5658
## 29	553.7138	541.8925	522.4762
## 30	530.6103	518.6782	502.3561
## 31	520.4992	509.3815	493.6204
## 32	501.5047	484.2408	470.0760
## 33	520.1988	512.4579	490.4354
## 34	553.5076	538.9828	522.4685
## 35	514.6818	503.4365	488.9047
## 36	501.9168	495.9985	476.7275
## 37	504.2134	492.8696	479.1520
## 38	470.5544	463.4811	450.0556
## 39	462.2389	457.5044	435.2816
## 40	560.0260	542.6601	530.8731
## 41	506.0894	499.6761	479.9178
## 42	475.5688	468.6740	454.6166
## 43	532.5333	510.5720	492.3170
## 44	471.2023	469.7422	458.1068
## 45	463.4421	460.6823	439.9626
## 46	547.5019	531.2995	523.6476
## 47	556.7803	539.6496	522.9036
## 48	507.2024	496.1195	476.8526
## 49	525.6626	516.2798	503.8384
## 50	518.6626	502.6383	487.6355
## 51	448.7921	438.7853	420.2429
## 52	536.6674	517.4169	498.9565

53 525.0977 505.4648 487.5221
54 537.1695 517.2946 506.0485
55 510.4819 497.8196 482.5925
56 501.1951 494.0438 483.5648
57 539.9978 527.6001 514.0218
58 497.0962 479.8772 465.2994
59 506.1621 487.8323 473.4026
60 357.1735 376.9820 366.2216
61 409.5507 422.4417 412.3886
62 390.0036 408.4186 405.0924
63 472.1939 480.0289 465.2561
64 401.1345 417.0382 394.0485
65 419.1479 436.4536 419.0002
66 423.3124 430.4241 412.8071
67 431.4507 440.5435 413.0913
68 401.5537 415.6029 395.6416
69 471.7004 476.7122 469.7607
70 425.9473 441.0074 420.8209
71 395.5981 407.7041 401.3255
72 518.5279 526.9466 506.8744
73 463.7891 476.3631 465.4036
74 506.6984 512.7032 489.8522
75 397.1627 406.6116 391.1261
76 351.0149 372.1120 356.7031
77 414.6156 430.7841 412.8496
78 389.7495 404.7490 393.4171
79 420.0650 430.9549 411.3124
80 459.3831 472.1722 451.5357
81 392.1981 414.1234 384.8908
82 440.5404 445.1146 423.2153
83 409.9570 426.9797 408.5520
84 433.0603 447.5659 445.3987
85 394.4571 403.8597 386.3221
86 446.6804 457.8044 435.4335
87 402.8554 424.1804 405.6654
88 403.4491 422.4811 403.0150
89 382.5908 396.3144 380.5023
90 404.4803 420.8521 400.1452
91 393.9655 410.5628 394.4632
92 394.3123 413.9751 401.9072
93 416.9966 433.8475 420.8616
94 418.6449 433.3185 410.8841
95 370.4409 388.7329 371.0867
96 464.6012 471.9920 448.7169
97 401.5262 415.2680 413.5805
98 399.2265 414.1503 388.3004
99 401.4687 410.2615 382.2151
100 427.5027 442.5860 411.7362
101 390.3728 403.7380 378.9939
102 412.6606 426.9632 409.6133
103 428.0629 438.5853 423.0381
104 354.9557 379.8262 362.4436
105 427.3813 437.7096 415.3832
106 433.0702 449.2516 428.2476
107 405.4686 417.5730 396.1950
108 422.3831 437.9239 423.8830

```

## 109 392.8836 413.8633 391.5462
## 110 402.4036 409.1016 383.3418
## 111 345.6909 368.0014 348.7327
## 112 409.4503 426.5139 408.5941
## 113 420.0119 427.2919 412.6955
## 114 356.0233 378.1234 360.6336
## 115 407.4790 422.8078 402.5802
## 116 360.2481 388.4288 354.5378
## 117 379.5119 398.5953 376.8209
## 118 426.2835 445.4646 422.4710
## 119 397.1284 411.0899 407.6353
## 120 394.1082 408.9005 389.5430
## 121 378.5594 388.2150 366.1625
## 122 431.4316 437.2108 403.3731
## 123 447.3663 462.1824 448.9274
## 124 444.7898 455.0799 441.6922
## 125 394.3653 406.6822 385.7113
## 126 395.8515 411.3754 387.1218
## 127 403.6135 418.2404 396.4003
## 128 395.9971 418.1451 401.9250
## 129 410.6322 431.2259 408.3866
## 130 388.2725 401.8026 393.6047
## 131 410.8486 421.9899 424.7144
## 132 402.9155 415.4112 424.7726
## 133 389.7411 407.1658 417.7586
## 134 388.6026 403.7133 414.9840
## 135 381.2785 398.1316 404.5223
## 136 387.0876 401.5680 412.2952
## 137 381.0645 394.8671 407.8336
## 138 408.5858 427.9083 442.6210
## 139 443.1478 455.6126 467.0303
## 140 434.3035 450.3753 459.9607
## 141 446.2154 454.7994 463.2508
## 142 449.5474 453.6174 463.8221
## 143 475.6079 486.2272 496.5858
## 144 467.9095 477.1631 486.6492
## 145 371.3841 380.0837 396.9502
## 146 426.9017 437.4038 446.8581
## 147 440.5601 453.1371 473.3040
## 148 400.0212 411.4512 433.8305
## 149 415.4188 428.0160 450.2810
## 150 399.4627 412.6230 433.4683
## 151 434.4816 446.4698 466.3804
## 152 375.1193 390.8971 415.3578
## 153 415.6230 435.4275 453.3328
## 154 409.1274 420.8840 444.3528
## 155 373.3894 392.9420 404.8982
## 156 415.0303 428.1483 450.7815
## 157 443.1879 453.9999 478.4616
## 158 395.0321 411.1776 429.1200
## 159 480.0250 490.7730 519.6569
## 160 424.5898 438.4916 464.7464
## 161 360.0282 379.2300 398.0606
## 162 479.0217 485.7127 499.3001
## 163 438.4481 451.8037 460.8922
## 164 421.5246 430.0320 441.3956

```

```
## 165 436.8208 448.6559 471.3945
## 166 450.0290 464.5545 481.5863
## 167 440.0289 450.7550 477.0525
## 168 382.8376 394.8556 418.8063
## 169 456.3221 465.1668 485.8843
## 170 450.1985 458.4540 485.7063
## 171 367.4596 384.1405 395.4190
## 172 370.0040 387.5017 410.8073
## 173 466.6196 472.2427 497.4518
## 174 457.4812 462.1208 486.8278
## 175 437.6253 448.4148 468.7611
## 176 420.4683 428.4432 455.6936
## 177 426.7725 435.3541 456.1091
## 178 477.0003 487.2572 516.0440
```

```
# Dự đoán từng quan sát thuộc vào loại rượu nào
type <- apply(S, 1, which.max)
type
```

```
## [1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [38] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [75] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [112] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [149] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
```

```
# Độ chính xác khi sử dụng quy tắc phân biệt Gauss để phân biệt loại rượu
true_type <- wine$Type
```

```
# Tính accuracy
accuracy <- mean(type == true_type)
accuracy
```

```
## [1] 1
```

```
library(MASS)
wines <- wine
wines$Type <- as.numeric(wines$Type)
ld <- lda(Type ~ ., data = wines)
ld # Tỷ lệ / xs thực; gttb mỗi nhóm; hệ số phân biệt tuyến tính: LD1, LD2
```

```
## Call:
## lda(Type ~ ., data = wines)
##
## Prior probabilities of groups:
##      1      2      3
## 0.3314607 0.3988764 0.2696629
##
## Group means:
##      Alcohol      Malic      Ash Alkalinity Magnesium Phenols Flavanoids
## 1 13.74475 2.010678 2.455593 17.03729 106.3390 2.840169 2.9823729
## 2 12.27873 1.932676 2.244789 20.23803 94.5493 2.258873 2.0808451
## 3 13.15375 3.333750 2.437083 21.41667 99.3125 1.678750 0.7814583
## Nonflavanoids Proanthocyanins Color Hue Dilution Proline
## 1 0.290000 1.899322 5.528305 1.0620339 3.157797 1115.7119
## 2 0.363662 1.630282 3.086620 1.0562817 2.785352 519.5070
## 3 0.447500 1.153542 7.396250 0.6827083 1.683542 629.8958
##
## Coefficients of linear discriminants:
##      LD1      LD2
```

```
## Alcohol      -0.403399781  0.8717930699
## Malic        0.165254596  0.3053797325
## Ash          -0.369075256  2.3458497486
## Alcalinity   0.154797889 -0.1463807654
## Magnesium    -0.002163496 -0.0004627565
## Phenols      0.618052068 -0.0322128171
## Flavanoids   -1.661191235 -0.4919980543
## Nonflavanoids -1.495818440 -1.6309537953
## Proanthocyanins 0.134092628 -0.3070875776
## Color        0.355055710  0.2532306865
## Hue          -0.818036073 -1.5156344987
## Dilution    -1.157559376  0.0511839665
## Proline      -0.002691206  0.0028529846
```

```
##
## Proportion of trace:
```

```
##      LD1      LD2
## 0.6875 0.3125
```

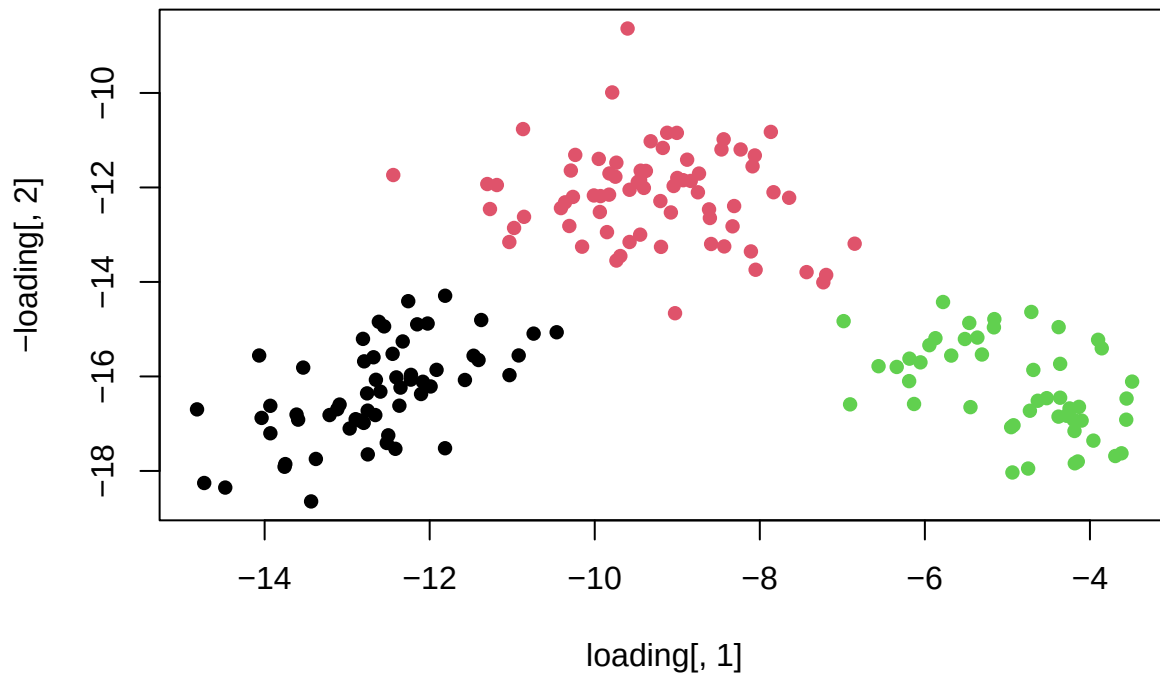
```
# theo các biến; % phân tách đạt được bởi từng hàm tuyến tính phân biệt
ld$scaling # Ma trận hệ số phân biệt tuyến tính
```

```
##              LD1              LD2
## Alcohol      -0.403399781  0.8717930699
## Malic        0.165254596  0.3053797325
## Ash          -0.369075256  2.3458497486
## Alcalinity   0.154797889 -0.1463807654
## Magnesium    -0.002163496 -0.0004627565
## Phenols      0.618052068 -0.0322128171
## Flavanoids   -1.661191235 -0.4919980543
## Nonflavanoids -1.495818440 -1.6309537953
## Proanthocyanins 0.134092628 -0.3070875776
## Color        0.355055710  0.2532306865
## Hue          -0.818036073 -1.5156344987
## Dilution    -1.157559376  0.0511839665
## Proline      -0.002691206  0.0028529846
```

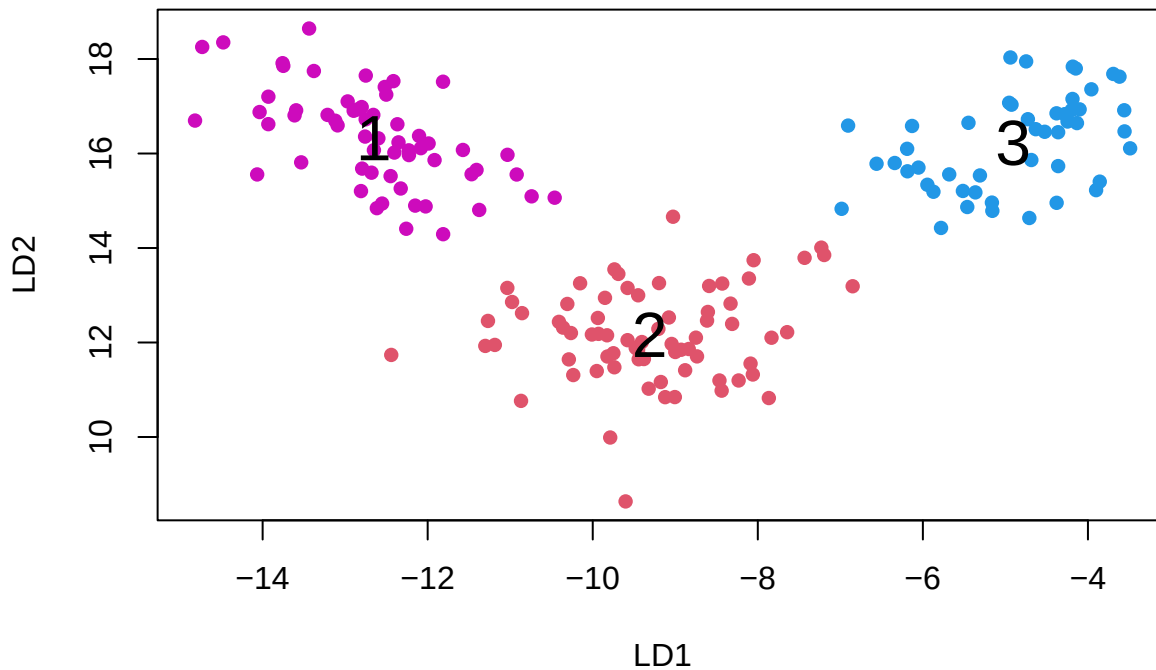
```
loading <- as.matrix(wines[, 2:14]) %*% ld$scaling
# loading <- predict(ld)$x # nolint
loading[1:2, ]
```

```
##              LD1              LD2
## [1,] -13.93103 16.62134
## [2,] -13.53274 15.81262
```

```
plot(loading[,1], -loading[,2], col = wines$Type, pch = 16)
```



```
plot(loading, col = c(6, 2, 4)[wines[, 1]],
     pch = 16, cex = 1.0, xlab = "LD1", ylab = "LD2")
for (i in 1:3) {
  centx <- mean(loading[wines[, 1] == i, ][, 1])
  centy <- mean(loading[wines[, 1] == i, ][, 2])
  text(centx, centy, i, cex = 2)
}
```

```
predict(ld)$posterior # XS dự đoán
```

##	1	2	3
## 1	1.000000e+00	3.261633e-09	3.641123e-18
## 2	9.999996e-01	3.583115e-07	8.733373e-17
## 3	9.999977e-01	2.321357e-06	7.823824e-14
## 4	1.000000e+00	3.726442e-12	1.334086e-16
## 5	9.251179e-01	7.488190e-02	2.171038e-07
## 6	1.000000e+00	3.509635e-11	1.291915e-17
## 7	1.000000e+00	2.700215e-11	1.200162e-17
## 8	1.000000e+00	1.905769e-10	2.304906e-16
## 9	9.999999e-01	6.006630e-08	2.436593e-15
## 10	9.999990e-01	9.839361e-07	1.156433e-13
## 11	1.000000e+00	7.882912e-10	1.560206e-18
## 12	9.999999e-01	1.013615e-07	6.790111e-14
## 13	1.000000e+00	3.178992e-08	1.063435e-14
## 14	1.000000e+00	1.223118e-10	3.713231e-21
## 15	1.000000e+00	2.485909e-13	6.100898e-21
## 16	1.000000e+00	1.166982e-08	4.124099e-13
## 17	1.000000e+00	1.376802e-08	1.859659e-13
## 18	9.999903e-01	9.693950e-06	2.068110e-12
## 19	1.000000e+00	3.895138e-13	4.347985e-20
## 20	9.999994e-01	6.131886e-07	6.646280e-13
## 21	9.999492e-01	5.075849e-05	2.562782e-14
## 22	9.935875e-01	6.412399e-03	5.452579e-08
## 23	9.999998e-01	1.748064e-07	1.444852e-18
## 24	9.997980e-01	2.019753e-04	1.062379e-12
## 25	9.996417e-01	3.582615e-04	1.915839e-13
## 26	9.688940e-01	3.110601e-02	1.717736e-09

```

## 27 1.000000e+00 1.588585e-08 9.262203e-16
## 28 9.999349e-01 6.506220e-05 2.377313e-11
## 29 9.999927e-01 7.346039e-06 2.714336e-14
## 30 9.999934e-01 6.575811e-06 5.362422e-13
## 31 9.999852e-01 1.484746e-05 2.121715e-12
## 32 1.000000e+00 3.179670e-08 2.242335e-14
## 33 9.995655e-01 4.344811e-04 1.185090e-13
## 34 9.999995e-01 4.919632e-07 3.310376e-14
## 35 9.999869e-01 1.306947e-05 6.385151e-12
## 36 9.973176e-01 2.682405e-03 1.146146e-11
## 37 9.999882e-01 1.184203e-05 1.306068e-11
## 38 9.991533e-01 8.467217e-04 1.250642e-09
## 39 9.912901e-01 8.709907e-03 1.944354e-12
## 40 1.000000e+00 2.871314e-08 2.182948e-13
## 41 9.983632e-01 1.636779e-03 4.296391e-12
## 42 9.989880e-01 1.011957e-03 7.945497e-10
## 43 1.000000e+00 2.899414e-10 3.422129e-18
## 44 8.115443e-01 1.884540e-01 1.667243e-06
## 45 9.404681e-01 5.953191e-02 5.974783e-11
## 46 9.999999e-01 9.191235e-08 4.367056e-11
## 47 1.000000e+00 3.633049e-08 1.938931e-15
## 48 9.999846e-01 1.537223e-05 6.595729e-14
## 49 9.999158e-01 8.415562e-05 3.325537e-10
## 50 9.999999e-01 1.098285e-07 3.350431e-14
## 51 9.999549e-01 4.508965e-05 3.992197e-13
## 52 1.000000e+00 4.361162e-09 4.191204e-17
## 53 1.000000e+00 2.975169e-09 4.798328e-17
## 54 1.000000e+00 2.335804e-09 3.050225e-14
## 55 9.999968e-01 3.168385e-06 7.722646e-13
## 56 9.992167e-01 7.832445e-04 2.202455e-08
## 57 9.999959e-01 4.127994e-06 5.233342e-12
## 58 1.000000e+00 3.325554e-08 1.551762e-14
## 59 1.000000e+00 1.095210e-08 5.925886e-15
## 60 2.496185e-09 9.999788e-01 2.122437e-05
## 61 2.520506e-06 9.999544e-01 4.304963e-05
## 62 9.708421e-09 9.653154e-01 3.468461e-02
## 63 3.954589e-04 9.996042e-01 3.837475e-07
## 64 1.239069e-07 9.999999e-01 1.036785e-10
## 65 3.049453e-08 9.999999e-01 2.630717e-08
## 66 8.148179e-04 9.991852e-01 2.231901e-08
## 67 1.124532e-04 9.998875e-01 1.195564e-12
## 68 7.916083e-07 9.999992e-01 2.142607e-09
## 69 6.608591e-03 9.924414e-01 9.499911e-04
## 70 2.880586e-07 9.999997e-01 1.710365e-09
## 71 5.517252e-06 9.982998e-01 1.694708e-03
## 72 2.206427e-04 9.997794e-01 1.917175e-09
## 73 3.460675e-06 9.999791e-01 1.739046e-05
## 74 2.460929e-03 9.975391e-01 1.188193e-10
## 75 7.877233e-05 9.999210e-01 1.882415e-07
## 76 6.880809e-10 9.999998e-01 2.032388e-07
## 77 9.508745e-08 9.999999e-01 1.626127e-08
## 78 3.060615e-07 9.999877e-01 1.198380e-05
## 79 1.864429e-05 9.999814e-01 2.946738e-09
## 80 2.790988e-06 9.999972e-01 1.090616e-09
## 81 3.005681e-10 1.000000e+00 2.015746e-13
## 82 1.020957e-02 9.897904e-01 3.053515e-10

```

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## 83 4.047302e-08 9.999999e-01 9.930801e-09
## 84 4.499907e-07 8.972724e-01 1.027272e-01
## 85 8.249501e-05 9.999175e-01 2.417933e-08
## 86 1.475382e-05 9.999852e-01 1.924970e-10
## 87 5.478239e-10 1.000000e+00 9.099192e-09
## 88 5.426103e-09 1.000000e+00 3.515237e-09
## 89 1.096344e-06 9.999988e-01 1.358075e-07
## 90 7.759467e-08 9.999999e-01 1.016527e-09
## 91 6.193200e-08 9.999998e-01 1.018648e-07
## 92 2.887574e-09 9.999943e-01 5.740564e-06
## 93 4.805769e-08 9.999977e-01 2.292493e-06
## 94 4.239848e-07 9.999996e-01 1.806643e-10
## 95 1.137306e-08 1.000000e+00 2.169411e-08
## 96 6.165573e-04 9.993834e-01 7.789350e-11
## 97 9.084754e-07 8.438891e-01 1.561100e-01
## 98 3.301184e-07 9.999997e-01 5.936754e-12
## 99 1.517906e-04 9.998482e-01 6.599749e-13
## 100 2.814446e-07 9.999997e-01 4.000107e-14
## 101 1.568781e-06 9.999984e-01 1.793826e-11
## 102 6.144220e-07 9.999994e-01 2.917636e-08
## 103 2.692405e-05 9.999729e-01 1.769766e-07
## 104 1.580737e-11 1.000000e+00 2.823873e-08
## 105 3.269330e-05 9.999673e-01 2.012488e-10
## 106 9.387183e-08 9.999999e-01 7.552388e-10
## 107 5.534996e-06 9.999945e-01 5.195827e-10
## 108 1.781183e-07 9.999990e-01 7.981622e-07
## 109 7.737830e-10 1.000000e+00 2.031331e-10
## 110 1.231762e-03 9.987682e-01 6.488025e-12
## 111 2.044858e-10 1.000000e+00 4.282228e-09
## 112 3.885179e-08 9.999999e-01 1.650185e-08
## 113 6.886971e-04 9.993108e-01 4.576745e-07
## 114 2.523764e-10 1.000000e+00 2.536709e-08
## 115 2.201920e-07 9.999998e-01 1.641537e-09
## 116 5.771512e-13 1.000000e+00 1.911248e-15
## 117 5.154331e-09 1.000000e+00 3.495310e-10
## 118 4.675000e-09 1.000000e+00 1.032802e-10
## 119 8.376742e-07 9.693670e-01 3.063215e-02
## 120 3.765197e-07 9.999996e-01 3.918935e-09
## 121 6.406351e-05 9.999359e-01 2.646662e-10
## 122 3.081851e-03 9.969181e-01 2.009740e-15
## 123 3.676696e-07 9.999979e-01 1.751668e-06
## 124 3.396939e-05 9.999645e-01 1.533926e-06
## 125 4.475589e-06 9.999955e-01 7.806592e-10
## 126 1.811570e-07 9.999998e-01 2.929478e-11
## 127 4.442480e-07 9.999996e-01 3.273236e-10
## 128 2.405754e-10 9.999999e-01 9.030136e-08
## 129 1.138345e-09 1.000000e+00 1.205023e-10
## 130 1.329868e-06 9.997235e-01 2.751561e-04
## 131 8.923808e-07 6.153941e-02 9.384597e-01
## 132 3.217713e-10 8.597404e-05 9.999140e-01
## 133 6.794305e-13 2.509693e-05 9.999749e-01
## 134 3.488842e-12 1.273963e-05 9.999873e-01
## 135 8.028169e-11 1.674312e-03 9.983257e-01
## 136 1.128387e-11 2.193992e-05 9.999781e-01
## 137 2.367710e-12 2.337355e-06 9.999977e-01
## 138 1.654617e-15 4.076869e-07 9.999996e-01

```

```
## 139 4.245868e-11 1.099872e-05 9.999890e-01
## 140 7.197671e-12 6.872235e-05 9.999313e-01
## 141 3.995118e-08 2.135381e-04 9.997864e-01
## 142 6.317509e-07 3.699549e-05 9.999624e-01
## 143 7.751488e-10 3.171845e-05 9.999683e-01
## 144 7.268153e-09 7.589190e-05 9.999241e-01
## 145 7.884375e-12 4.731136e-08 1.000000e+00
## 146 2.152927e-09 7.834667e-05 9.999217e-01
## 147 6.018765e-15 1.744298e-09 1.000000e+00
## 148 2.073985e-15 1.909049e-10 1.000000e+00
## 149 7.236065e-16 2.140096e-10 1.000000e+00
## 150 1.704439e-15 8.851366e-10 1.000000e+00
## 151 1.401212e-14 2.253888e-09 1.000000e+00
## 152 3.346918e-18 2.381377e-11 1.000000e+00
## 153 4.195729e-17 1.674300e-08 1.000000e+00
## 154 5.032898e-16 6.421173e-11 1.000000e+00
## 155 2.069701e-14 6.419086e-06 9.999936e-01
## 156 2.974894e-16 1.480965e-10 1.000000e+00
## 157 4.795601e-16 2.379184e-11 1.000000e+00
## 158 1.569685e-15 1.613223e-08 1.000000e+00
## 159 6.138686e-18 2.856824e-13 1.000000e+00
## 160 3.632449e-18 3.959640e-12 1.000000e+00
## 161 3.039034e-17 6.637100e-09 1.000000e+00
## 162 1.560237e-09 1.256253e-06 9.999987e-01
## 163 1.789043e-10 1.129453e-04 9.998871e-01
## 164 2.344863e-09 1.161007e-05 9.999884e-01
## 165 9.656814e-16 1.332647e-10 1.000000e+00
## 166 1.971666e-14 4.010267e-08 1.000000e+00
## 167 8.333929e-17 3.794619e-12 1.000000e+00
## 168 2.393434e-16 3.966000e-11 1.000000e+00
## 169 1.449826e-13 1.005755e-09 1.000000e+00
## 170 3.794601e-16 1.460398e-12 1.000000e+00
## 171 7.200244e-13 1.264093e-05 9.999874e-01
## 172 1.902559e-18 7.559555e-11 1.000000e+00
## 173 4.071087e-14 1.126758e-11 1.000000e+00
## 174 1.798520e-13 1.861473e-11 1.000000e+00
## 175 3.005318e-14 1.457729e-09 1.000000e+00
## 176 5.033243e-16 1.463164e-12 1.000000e+00
## 177 1.816681e-13 9.687823e-10 1.000000e+00
## 178 1.105427e-17 3.148141e-13 1.000000e+00
```

```
predict(ld)$posterior[100, ] # XS dự đoán dựa trên hàm pb tuyến tính  $S_i(U)$ 
```

```
##           1           2           3
## 2.814446e-07 9.999997e-01 4.000107e-14
```

```
# XS thuộc nhóm  $i = \exp(S_i(U)) / \sum(S_j(U))$ 
```

```
data <- wine[100, -1]
```

```
predict(ld, data)$class # Dự đoán 1 quan sát thuộc nhóm nào
```

```
## [1] 2
```

```
## Levels: 1 2 3
```